Predicting Coronary Heart Disease from BRFSS 2015 with Plain SGD and L2-Regularised Logistic Regression

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Machine Learning Course — Project 1

***Abstract*—We predict reported coronary heart disease, myocardial infarction, or angina from the 2015 BRFSS (328,135 labeled; 8.83% positive; 44.8% missing). We use plain single-sample SGD logistic regression with an all-weight L2 penalty. On the seed-42 training-side partition (also used earlier), $\lambda = 10^{-5}$ had the best calibration F1 and a 27% smaller norm than $\lambda = 0$. Full-data 5-fold CV gives 0.4303 ± 0.0060 F1, 0.8622 AUROC and 0.4188 PR-AUC; unregularized encoded gives 0.4315 ± 0.0053 F1. The frozen threshold is 0.2148; final predictions are 17,310 positive.**

I. Introduction and Data

Coronary heart disease is a leading cause of death, and survey data can help identify respondents with a reported history of heart disease. We use the 2015 BRFSS [1]; the target `_MICHD` is positive when a respondent reports that a health professional told them they had coronary heart disease, a myocardial infarction, or angina. The training set has 328,135 labeled rows (28,975 positive, 8.83%), 321 raw features, substantial missingness and coded categorical variables. The test set has 109,379 rows.

II. Preprocessing and Feature Representation

For every fitting portion, medians, means, standard deviations and category sets are learned from that portion only and applied unchanged to held-out or test data. We remove 14 identifiers and survey-design fields (interview date components, record number, disposition code, sampling unit, stratum and survey weights), remove constant columns, median-impute and standardise numeric features, and one-hot encode features with at most 15 distinct finite training values. We use $K - 1$ dummies, the most frequent value as reference, and an explicit missing category. Designs have 1,129–1,130 columns on the 262,508-row training portions and 1,120 on all data.

A header-level check confirmed that `_MICHD` is the separate target and that `CVDINFR4` and `CVDCRHD4`, direct source variables for its definition, are absent from the predictors; this does not prove the absence of every derived relationship. We retained `CVDSTRK3`, which records stroke history rather than the target condition.

III. Six Required Course Algorithms

`implementations.py` contains the six required functions: `mean_squared_error_gd`, `mean_squared_error_sgd`, `least_squares`, `ridge_regression`, `logistic_regression` and `reg_logistic_regression`. Their signatures and behavior are unchanged, and they pass all 25 public tests. The final model and the experiments use the shared plain-SGD routine in `pipeline.py`; `run.py` is the final prediction entry point.

IV. Plain Stochastic Gradient Descent

For labels $y_i \in \{0, 1\}$ and logits $z_i = x_i^\top w$, the mean logistic loss is

$$L(w) = \frac{1}{N} \sum_{i=1}^{N} \left[\log(1 + e^{z_i}) - y_i z_i \right]. \tag{1}$$

For one example, the update is

$$w \leftarrow w - \eta_t \left(\sigma(x_i^\top w) - y_i \right) x_i, \tag{2}$$

where $\sigma(z) = 1/(1 + e^{-z})$ and t is the epoch index. We use the inverse schedule $\eta_t = 0.01/(1+0.1(t-1))$ for 90 epochs, reshuffling each epoch with seed 2027. There are no moment estimates or per-parameter scaling. We use a stable sigmoid and log loss and compute the full loss only at checkpoints.

We verified the analytic gradient against central finite differences for the unregularized and L2 objectives, the average of per-sample gradients against the full-data gradient, a manual update, schedule values, seed-reproducible shuffling, finite losses, and the absence of adaptive state. The loss decreased on a controlled synthetic problem. Because SGD is noisy, we use a fixed epoch cap rather than validation early stopping.

V. L2 Regularisation

The penalized objective is $J(w) = L(w) + \lambda \sum_{j=0}^{d} w_j^2$, so

$$w \leftarrow w - \eta_t \left[\left(\sigma(x_i^\top w) - y_i \right) x_i + 2\lambda w \right]. \tag{3}$$

The penalty includes the intercept, matching the course convention. We studied $\lambda \in \{0, 10^{-5}, 10^{-4}, 10^{-3}\}$ on the seed-42 stratified 80/20 partition of all labeled data: 262,508 fit and 65,627 calibration rows. This partition was also used during earlier development. The fitting rows train each λ; the calibration rows select the threshold and supply the F1 values, so this study describes model selection and regularization behavior, not independent generalization. It is not the five-fold CV; the same SGD configuration and epochs were used for every λ.

TABLE I
L2 STUDY ON THE SEED-42 TRAINING-SIDE 80/20 PARTITION (262,508 FIT, 65,627 CALIBRATION), ALSO USED EARLIER; F1 USES A CALIBRATION-OPTIMIZED THRESHOLD, NOT INDEPENDENT VALIDATION. Δ IS CALIBRATION MINUS TRAINING LOSS.

λ	F1	AUROC	$\|w\|_2$	Δ
0	0.4368	0.8622	10.537	$+0.00219$
10^{-5}	**0.4374**	**0.8625**	7.663	$+0.00182$
10^{-4}	0.4360	0.8623	4.180	$+0.00109$
10^{-3}	0.4284	0.8581	2.060	$+0.00062$

TABLE II
FULL-DATA STRATIFIED FIVE-FOLD CV ON ALL 328,135 LABELED EXAMPLES. F1 IS MEAN $\pm$ SAMPLE SD OVER FIVE HELD-OUT FOLDS; AUROC AND PR-AUC ARE FOLD MEANS. THRESHOLDS ARE SELECTED ON INNER DATA ONLY.

Model	F1 (mean $\pm$ SD)	AUROC	PR-AUC
Majority (-1)	0.0000 ± 0.0000	0.5000	0.0883
Ridge (encoded)	0.4272 ± 0.0027	0.8583	0.4161
Logistic (numeric)	0.4022 ± 0.0036	0.8411	0.3502
Logistic (encoded)	0.4315 ± 0.0053	0.8619	0.4188
L2 logistic (encoded)	$\mathbf{0.4303 \pm 0.0060}$	**0.8622**	**0.4188**

The weakly regularized model achieved a slightly lower calibration logistic loss (0.22021 versus 0.22037) and a 27% smaller norm. $\lambda = 10^{-4}$ shrinks the norm by 60% at a cost of 0.0014 F1; $\lambda = 10^{-3}$ underfits (F1 0.4284, AUROC 0.8581). The training–calibration gap narrows with stronger penalties. A smaller norm alone is not evidence of better generalization, and the CV shows no uniform F1 advantage for the regularized model; $\lambda = 10^{-5}$ is retained because the calibration study supports it and coefficients are controlled.

VI. FULL-DATA STRATIFIED FIVE-FOLD CROSS-VALIDATION

We run one stratified five-fold CV on all 328,135 labeled examples (28,975 positives, 299,160 negatives). Each fold holds out exactly 65,627 rows: 5,795 positives and 59,832 negatives. Every row appears in exactly one evaluation fold and four training sets; the fold seed is 2028 and assignments are saved.

Each 262,508-row outer training portion is split, stratified, into 210,006 fitting and 52,502 inner calibration rows. Preprocessing is fitted only on the fitting rows. The threshold is chosen on inner calibration data to maximize positive-class F1 and frozen; the model and preprocessing are refitted on the complete outer training portion, and the threshold is applied once to the held-out fold. No held-out label influences its own preprocessing, threshold, or fit. All models use the same folds.

Hyperparameters, λ, and the final threshold were frozen before this CV; all labeled rows had participated in earlier training-side development, so the CV is a retrospective evaluation of a frozen procedure, not an untouched independent test. The folds are correlated and their SD is descriptive.

The unregularized encoded model has the largest mean F1 (0.4315); the selected L2 model is 0.0012 lower. The paired difference, F1(L2) $-$ F1(unregularized), is -0.0012 ± 0.0029 and changes sign across folds. L2 has slightly higher AUROC and the same PR-AUC. We do not claim a significant or uniform advantage; L2 is retained because the separate calibration study selected it, it controls coefficient magnitude, and full-data F1 is comparable.

The encoded model's paired mean F1 advantage over numeric codes is $+0.0293 \pm 0.0043$ (encoded minus numeric); over ridge it is $+0.0043 \pm 0.0051$, and ridge has lower mean AUROC. These differences are descriptive, not significance tests.

VII. THRESHOLD SELECTION AND FINAL MODEL

The final threshold is 0.21475973018819802, selected to maximize positive-class F1 on an inner split of the 262,508-row seed-42 training portion (210,006 fit, 52,502 calibration, inner seed 123; inner F1 0.4336) and frozen before the all-data fit. Each CV fold selects its own threshold (0.1538–0.2427). The regularization study instead fits on all 262,508 rows and optimizes its own threshold (about 0.1706, F1 about 0.4374); applying the frozen threshold to those scores gives F1 about 0.4255. These are different, non-independent procedures.

The final L2 model is trained on all 328,135 labeled rows with the frozen configuration and applied to the 109,379 test rows. It produces 17,310 positive predictions (15.83%). Test labels are unavailable, so no test performance is reported. The final training loss is 0.2189.

VIII. DISCUSSION AND LIMITATIONS

The model is linear and cannot capture non-linear interactions; no resampling or class weighting was used. The full-data CV is retrospective because all labeled rows informed earlier preprocessing, hyperparameter, and threshold choices, and its folds are correlated. The 90-epoch cap and schedule were frozen before CV; we do not claim convergence. The L2 experiment measures coefficient shrinkage and calibration-partition loss, not dedicated probability calibration. Associations are not causal, the threshold is a point estimate, and test labels are unavailable.

IX. CONCLUSION

A NumPy-only pipeline combines one-hot encoding, verified preprocessing, plain single-sample SGD, an all-weight L2 penalty, and leakage-free fold-level threshold selection. Full-data five-fold CV gives 0.4303 ± 0.0060 F1, 0.8622 AUROC and 0.4188 PR-AUC; the unregularized encoded model is 0.0012 higher, a small and mixed difference. The final model produces 17,310 positive test predictions.

REFERENCES

[1] Centers for Disease Control and Prevention. *2015 BRFSS Survey Data and Documentation*. https://www.cdc.gov/brfss/annual_data/annual_2015.html, 2016.